

freeiv: Instrument-free estimation of a linear structural model with an endogenous regressor

Abdelkrim Araar
Université Laval and Partnership for Economic Policy
Québec, Canada
aabdd@ecn.ulaval.ca

Abstract. `freeiv` estimates the coefficient on an endogenous regressor when no external instrument is available, in the linear triangular model in which a single latent confounder generates the endogeneity. Under scale consistency — the confounder moves the outcome directly by exactly as much as it moves it through the regressor — the second moments of the data leave one number free, and the coefficient lies between half the least-squares slope and that slope. The interval assumes nothing more, and it is exactly the set of values for which the implied variances are non-negative. The routes of the command impose the same linear structure on the moments of the next orders: the third order identifies the coefficient (the quadratic moment estimator, the default, and a closed form for symmetric noise), and the fourth over-identifies it (a joint and a profiled generalized method of moments estimator, each with a Hansen J statistic). A second indicator of the same confounder frees the confounder's direct loading and turns scale consistency into a test. The estimators of Lewbel, Schennach, and Zhang (2024, *Journal of Business and Economic Statistics* 42: 14–25) and of Lewbel (2012, *Journal of Business and Economic Statistics* 30: 67–80), the Gaussian copula, the rank control function, and the coefficient-stability adjustment of Oster (2019, *Journal of Business and Economic Statistics* 37: 187–204) are computed on the same data and judged by the same interval. Standard errors come from the influence functions of the moments, with sampling weights in sandwich form and linearization over a survey design. Companion commands report what the data can carry before any estimation, what a value of the coefficient implies about the unobservables, and tests of endogeneity and of the agreement between routes.

Keywords: `freeiv`, `freeivmenu`, `freeivdiag`, `freeivtest`, `freeivreport`, endogeneity, latent confounder, scale consistency, partial identification, higher-order moments, instrument-free identification, survey data

1 Introduction

The applied researcher who suspects that a regressor is endogenous and has no credible instrument has, in practice, three options. Report the least-squares coefficient and discuss the direction of the bias. Find an instrument whose exclusion restriction is defensible, which is rarely possible and never testable. Or turn to one of the instrument-free estimators that identify the coefficient from features of the data other than an exclusion restriction: heteroskedasticity, higher-order moments, a functional form for the dependence between the disturbances.

The third option has produced a substantial literature, and it has an uncomfortable property. Each of these estimators identifies the parameter inside its own maintained model, and the maintained models differ. Applied to the same dataset they give different numbers, and the difference is not sampling noise: it is the disagreement between the assumptions. A reader shown one of them has no way to tell whether the assumption behind it holds.

freeiv is built around that difficulty rather than around any one estimator. It takes one linear structure seriously enough to ask what can be said before any distributional assumption is added, reports the answer as an interval, and then shows every point estimator against that interval together with the statistic that says whether its own assumption is carried by these data. An estimate that falls outside the interval is not a large estimate; it is an infeasible one, implying a negative variance under the maintained model, and the command says so instead of printing it without comment.

The output is meant to be read in four steps. **freeivmenu**, before any estimation, says which routes the data can carry. The interval comes next: it holds whatever route is chosen. The routes follow, each against the interval. Last come the diagnostics that say whether the point a route returns can be read: the precision of the third-order signal, the share of the interval that the fourth-order moments rule out, the tests of the over-identifying restrictions.

Section 2 presents the models: what each route assumes, what it identifies, how it fails and what announces the failure. There are no proofs here — they are in the papers cited at each point — and the notation is kept to what a practitioner needs to decide whether a route applies to the problem at hand. Section 3 presents the command: syntax, options, the output, standard errors, weights and survey data, the bootstrap, stored results, and the companion commands. Section 4 works through examples on simulated data where the truth is known and on real data where it is not, and Section 5 discusses what the approach can and cannot do.

2 The models

2.1 The setting

Throughout, Y_1 is the outcome, Y_2 the endogenous regressor, X a vector of exogenous controls including a constant, and γ_1 the coefficient of interest:

$$Y_1 = X'\beta_1 + \gamma_1 Y_2 + \varepsilon_1, \quad \varepsilon_1 = \alpha_1 U + V_1, \quad (1)$$

$$Y_2 = X'\beta_2 + \varepsilon_2, \quad \varepsilon_2 = \alpha_2 U + V_2. \quad (2)$$

The endogeneity has one source: a latent factor U that enters both equations. U , V_1 and V_2 are independent of one another and of X . Least squares is biased because U enters both equations, and its bias is proportional to $\alpha_1 \alpha_2$: a confounder that moved the regressor alone would be harmless (Araar 2026d). Nothing so far is identified: γ_1 , α_1 , α_2 and the distribution of U can trade off against one another without changing what is observed.

2.2 Scale consistency

The restriction that closes the model is *linear scale consistency*:

$$\alpha_1 = \gamma_1 \alpha_2. \quad (3)$$

In words: the confounder moves the outcome directly by exactly as much as it moves it through the regressor. One unit of U shifts Y_2 by α_2 , that shift carries into Y_1 through γ_1 , and (3) says the direct effect of U on Y_1 has that same size. It is a substantive restriction, plausible in some applications and not in others.

It has one consequence a practitioner should keep in view. Under (3), least squares can only exaggerate the size of γ_1 , never understate it. Any value beyond the least-squares slope — above it, for a positive effect — is therefore incompatible with the model: an instrumental-variable estimate larger than least squares, or the truth itself when the regressor is measured with error and least squares is attenuated. Model A below maintains (3); the two-indicator model of Section 2.5 frees α_1 and tests it.

Write $\hat{\xi}$ and $\hat{\varepsilon}_2$ for the residuals of Y_1 and Y_2 after regressing each on X , and let

$$m_{jk} = E[\hat{\xi}^j \hat{\varepsilon}_2^k]$$

denote their moments. Every route below is a function of a handful of these. Two quantities recur:

$$\theta = \alpha_2^2 \text{Var}(U), \quad c = \frac{\theta}{\theta + \text{Var}(V_2)} \in [0, 1],$$

θ being the part of the first-stage residual variance due to the confounder and c its share.

2.3 One structure, one missing number

Under (3), $\hat{\xi}$ behaves as $2\gamma_1\alpha_2U + \gamma_1V_2 + V_1$, and the three second moments are

$$m_{02} = \theta + \sigma_{V_2}^2, \quad m_{11} = \gamma_1(m_{02} + \theta), \quad m_{20} = 4\gamma_1^2\theta + \gamma_1^2\sigma_{V_2}^2 + \sigma_{V_1}^2 :$$

three equations in four unknowns, $(\gamma_1, \theta, \sigma_{V_1}^2, \sigma_{V_2}^2)$. One number is missing. Take it to be c : the least-squares slope $\tilde{\gamma} = m_{11}/m_{02}$ then satisfies

$$\tilde{\gamma} = \gamma_1(1 + c), \quad (4)$$

and since $c \in [0, 1]$ by construction,

$$\gamma_1 \in [\tilde{\gamma}/2, \tilde{\gamma}] \quad \text{for } \gamma_1 > 0, \quad (5)$$

with the endpoints exchanged when $\gamma_1 < 0$. Three things follow, and they are the backbone of the command.

1. **The sign is identified.** Least squares cannot reverse the sign of the coefficient under this model.

2. **The direction of the correction is known.** Any consistent estimator must be *smaller in magnitude* than the least-squares coefficient, and no smaller than half of it.
3. **The interval is exactly the feasible set.** $c < 0$ or $c > 1$ means a negative variance. A candidate outside (5) is therefore not a large estimate but an infeasible one, and the command reports it as such.

The two ends are estimated, and each carries its standard error. Running from the lower end less 1.96 standard errors to the upper end plus 1.96 gives a conservative 95% confidence set for γ_1 ; Imbens and Manski (2004) give a tighter one. No distribution has been assumed. The interval fails only when the model fails — a regressor measured with error, a confounder whose direct effect exceeds or opposes the one it transmits — and nothing in model A can see that; model B can. This is why **freeiv** prints the interval on every call, whatever route is requested.

2.4 The structure, order after order

To move from the interval to a point, the missing number must be obtained. Every route of model A does it in the same way: it imposes the same linear structure on the moments of the next order. At order three, three more moments (m_{03}, m_{12}, m_{21}) bring two more unknowns, $A = \alpha_2^3 E[U^3]$, the confounder's part of the third moment of the first-stage residual, and $B = E[V_2^3]$, the rest: γ_1 is exactly identified. At order four, three more moments bring two more unknowns: one restriction is over-identifying, and the structure becomes testable. Table 1 sums up the staircase; it is the structure, not an assumption from outside it, that does the work at each step, and skewness is what makes the higher moments informative.

Table 1: The linear structure imposed order after order

moments	added	unknowns added	what the structure gives
order 2	m_{02}, m_{11}, m_{20}	$\gamma_1, \theta, \sigma_{V_1}^2, \sigma_{V_2}^2$	one short: the interval (bounds)
order 3	m_{03}, m_{12}, m_{21}	A, B	γ_1 exactly identified (qme ; hme if the noise is symmetric)
order 4	m_{04}, m_{13}, m_{22}	A_4, B_4	one over-identifying restriction, a J on 1 degree of freedom (gmm , pgmm)
two indicators	16, orders 2–3	12, α_1 free	four over-identifying restrictions; scale consistency tested (model B)

Order three: the quadratic moment estimator

The default route. Under (3) the three third-order cross-moments satisfy a quadratic in γ_1 ,

$$2m_{03}\gamma_1^2 - 3m_{12}\gamma_1 + m_{21} = 0, \quad (6)$$

whose roots are γ_1 and $\gamma_1(4A + B)/\{2(A + B)\}$, and whose discriminant is

$$D = 9m_{12}^2 - 8m_{03}m_{21} = \gamma_1^2(2A - B)^2.$$

The route keeps the root inside (5). It needs a third-order signal in the first-stage residual that is not split as $B = 2A$. A skewed confounder provides it; a skewed V_2 alone does too, the other root being then $\gamma_1/2$, below the interval. Araar (2026b) finds the route usable from a skewness of the first-stage residual of 0.5 and reliable from 1.

How it fails is printed, not hidden. Near $D = 0$ the two roots merge and the standard error, proportional to $1/\sqrt{D}$, grows. When D falls below zero, the vertex $3m_{12}/(4m_{03})$, where the roots merge, is returned and flagged; it coincides with γ_1 when $B = 2A$. A D below zero by more than its sampling noise cannot come from the model at all, and the output then says that the data reject it. When no root lies inside the interval, the model is refuted on these data, and the output says that too. `freeivmenu` prints the z of m_{03} and the z of D before any estimation. See Araar (2026b) and Araar (2026c).

`method(hme)` is the closed form $\frac{1}{2}(m_{30}/m_{03})^{1/3}$. It adds to the quadratic's requirement that V_1 and V_2 be symmetric: then $m_{30}/m_{03} = 8\gamma_1^3$. It is the cheapest route when it applies and the most dangerous when it does not, because with a skewed noise it still returns a plausible number. The output and `freeivmenu` read B at the retained value; $E[V_1^3]$ cannot be read. When m_{30}/m_{03} does not have the sign of $\tilde{\gamma}$, the route returns nothing (Araar 2026a).

Order four: two GMM routes and a test

Adding the fourth-order moments gives nine moment conditions in eight parameters, $(\gamma_1, \theta, \sigma_{V_1}^2, \sigma_{V_2}^2, A, B, A_4, B_4)$: one over-identifying restriction and a Hansen J on one degree of freedom (Araar 2026c).

`method(gmm)` minimizes the criterion *deterministically*, and the reason is worth a paragraph. Fix γ_1 and θ : the only nonlinearity left is the product $\theta\sigma_{V_2}^2$, and fixing θ removes it, so all nine residuals become linear in the six remaining parameters. The eight-parameter problem is therefore a two-dimensional surface in (γ_1, θ) with a bounded linear least-squares step at each node, and a surface can be gridded and refined. There is no seed and no starting value: the same data give the same number. Multistart local optimization over the eight parameters does not have that property: on six of the eight applications of Araar (2026c) it stops at a local minimum.

Because the criterion is often nearly flat, the route reports more than a point. Profiling J over γ_1 and collecting

$$\mathcal{C} = \{\gamma_1 \in [\tilde{\gamma}/2, \tilde{\gamma}] : J(\gamma_1) - \min J \leq 3.84\} \quad (7)$$

gives the region where the profiled J stays within 3.84 of its minimum. The command prints $\mathcal{C}$ and its share of the interval. **That share is the statistic to read first.** It rests on no standard error: when it is well below one, the moments of orders three and four narrow what the second order gives; when it fills the interval — as on seven of the eight applications of Araar (2026c) — they have added nothing to (5), the point estimate is the argmin of a flat surface, and the interval is what to report. $\mathcal{C}$ describes how flat the criterion is; it is not a confidence set with a guaranteed coverage under weak identification, which a test of the moment conditions at each fixed γ_1 against its own χ^2 law would give (Stock and Wright 2000). A minimum on the boundary of the parameter space gets no standard error, since the delta method has nothing to expand around there.

`method(pgmm)` fits the conditions of orders two and three exactly, which solves $\theta, \sigma_{V_1}^2, \sigma_{V_2}^2, A, B$ in closed form at each γ_1 and leaves four residuals in three unknowns: again one over-identifying restriction. It is a *different* estimator, not the same one solved differently. Its search is one-dimensional and cheap, so its J is printed under every route as the test at order four; and substituting A into its fifth residual returns $m_{21} - 3\gamma_1 m_{12} + 2\gamma_1^2 m_{03}$, the quadratic (6) itself. A wide gap between the two GMM signals a flat criterion, not an error in either (Araar 2026c, section 6.3).

What the J tests, and when it can test anything. Scale U so that $\hat{\varepsilon}_2 = U + V_2$; then $\hat{\xi} = \tau U + \gamma_1 V_2 + V_1$, and (3) is $\tau = 2\gamma_1$. The J tests that restriction jointly with the linearity of the confounder's effect; a rejection does not say which failed. And it cannot always test: with τ free, the Jacobian of the nine moments has determinant

$$-(\gamma_1 - \tau)^5 (B \kappa_U - A \kappa_{V_2}),$$

κ denoting excess kurtosis, so the restriction is testable everywhere except where the second factor vanishes — and a normal V_2 sits exactly there, since it makes both B and κ_{V_2} zero. `freeiv` reports that factor with a z , and says when a non-rejection carries no information because the design sits near that surface.

2.5 A second window on the confounder: two indicators

When a second endogenous variable loads on the same latent factor, the model becomes

$$Y_1 = X'\beta_1 + \gamma_2 Y_2 + \gamma_3 Y_3 + a_1 U + V_1, \quad (8)$$

$$Y_2 = X'\beta_2 + a_2 U + V_2, \quad Y_3 = X'\beta_3 + a_3 U + V_3, \quad (9)$$

and the situation changes in kind. Under one factor (U, V_1, V_2, V_3 independent), a skewed confounder, relevant indicators, and an indicator not caused by the regressor (Y_2 does not enter the equation of Y_3), Theorem 1 of Araar (2026d) gives γ_2, γ_3, a_1 , the loadings, $E[U^3]$ and the idiosyncratic variances in closed form, from the moments of orders two and three. The direct loading a_1 is *free*: scale consistency becomes a hypothesis the data can refute, and `freeiv` reports the test as $a_1 - (\gamma_2 a_2 + \gamma_3 a_3)$ with its standard error. A GMM on sixteen moments gives a J on four degrees of freedom

against the one-factor linear structure, and three estimates of a_2/a_3 that must agree under one factor give a second check. Araar (2026d) asks for a correlation between the two indicators with t above 10. The approach belongs to the identification of causal effects from proxies of an unmeasured confounder (Kuroki and Pearl 2014; Miao et al. 2018).

Y_3 is *not* an instrument: it is correlated with the very confounder it would be meant to purge. When Y_3 has no effect of its own on the outcome, using it as an instrument for Y_2 returns $\gamma_2 + a_1/a_2$ — twice γ_2 under scale consistency — and the command prints that value, so the gap is visible.

Three guards, those of Proposition 1 of Araar (2026d), protect the closed form. They fire when the moments imply something no one-factor structure can produce: the two third-order cross-moments of the indicators' residuals have opposite signs, so the confounder's skewness cannot orient the loadings; the product of the loadings would be negative; or an implied idiosyncratic variance is negative. When a guard fires no estimate is reported, the command ends with `r(498)`, and `e(guard)` records which one fired. A refusal is information about the candidate indicator, not a failure of the command.

2.6 The other maintained models, on the same data

Five further routes close the system with an assumption of their own, not with scale consistency. They are computed on the same data so that the interval judges them all: an estimate outside it implies a negative variance under scale consistency, and `freeivdiag` says which. Table 2 lists every route with what it adds, the statistic of `freeivmenu` that says whether the data carry it, and its references.

`lsz` is the same triangular system with the confounder's direct loading free, not scale consistency, identified by non-Gaussian errors through moments up to order five (Lewbel et al. 2024). With $p = (0, 1)$, as `trigmm` implements it (Lee et al. 2026), the system is exactly identified and solved as a root, and `freeiv` reproduces `trigmm`'s estimate within its stopping tolerance. Read `e(lsz_conv)` first: a solver that did not converge shows no standard error, and its point is not an estimate. Where the solver stops, `trigmm` with several starts remains the reference. The parameter spaces nest; the identifying assumptions do not, so confronting the route with the interval tests the scale-consistent model, not LSZ.

`lewbel12` is two-stage least squares with the instruments $(X - \bar{X})\hat{\varepsilon}_2$ (Lewbel 2012), as `ivreg2h` (Baum and Schaffer 2012) computes it. It needs the first-stage error to be heteroskedastic in X — `freeivmenu` prints the F — and X uncorrelated with $\varepsilon_1\varepsilon_2$; in the one-factor model the second holds when the heteroskedasticity comes from V_2 , not from U (Araar 2026a). `copula` adds the control function $\Phi^{-1}\{\bar{F}(Y_2)\}$ to the outcome equation, under a Gaussian copula between the regressor and the structural error and with a non-normal regressor (Park and Gupta 2012). `rank` applies the same idea to the ranks of the first-stage residual, which takes the controls out of the transformation (Breitung et al. 2024). `oster` is the bias-adjusted coefficient under proportional selection

Table 2: The routes, what each adds to the linear model, the signal that says whether the data carry it, and the references

route	adds to the model	signal	reference
bounds	nothing	—	Araar (2026a,c)
ols	nothing; the upper end	—	—
qme	a third-order signal, not on $B = 2A$	z of m_{03} , z of D	Araar (2026b,c)
hme	V_1 and V_2 symmetric	B	Araar (2026a)
gmm	the fourth-order identities	region share, J , factor	Araar (2026c)
pgmm	the same, orders 2 and 3 exact	J	Araar (2026c)
model B	a second indicator; α_1 free	t of the pair, guards	Araar (2026d)
lsz	α_1 free; non-Gaussian errors, moments to order 5	convergence, criterion	Lewbel et al. (2024); Lee et al. (2026)
lewbel12	heteroskedasticity in X ; $\text{Cov}(X, \varepsilon_1 \varepsilon_2) = 0$	F of ε_2^2 on X	Lewbel (2012); Baum and Schaffer (2012)
copula	a Gaussian copula; a non-normal Y_2	—	Park and Gupta (2012)
rank	a non-Gaussian first-stage residual	—	Breitung et al. (2024)
oster	proportional selection; δ and $R_{\max}$ assumed	—	Oster (2019); Oster (2013)

on observables and unobservables (Oster 2019), as **psacalc** (Oster 2013) computes it; δ and $R_{\max}$ are assumed, not estimated, so the honest report is a range.

3 The command

3.1 Syntax

One endogenous regressor:

```
freeiv depvar [indepvars](endogvar) [if][in][weight][, method(name)
delta(#) rmax(#) sign(#) vce(svy) level(#) noheader]
```

Two endogenous regressors sharing one latent confounder:

```
freeiv depvar [indepvars](endogvar1 endogvar2) [if][in][weight][, vce(svy)
level(#) noheader]
```

The second model is selected by the syntax alone — two variables inside the parentheses — and needs no method name. **pweights** and **aweight**s are allowed; under **vce(svy)** the weights are those of **svyset**. *indepvars* may contain factor variables, and the estimate is the one obtained from indicators built by hand. The **bootstrap** and **jackknife** prefixes are allowed; the **svy** prefix is not, and **vce(svy)** takes its place.

3.2 Options

`method(name)` selects the route of model A: `bounds`, `ols`, `qme` (the default), `hme`, `gmm`, `pgmm`, `lsz`, `lewbel12`, `copula`, `rank`, `oster`, or `all`. Every route estimates the same γ_1 ; what separates them is the assumption each adds. `method(all)` prints every route against the interval, which is how they are meant to be read.

`delta(#)` and `rmax(#)` act on `method(oster)`: Oster's ratio of selection on unobservables to selection on observables, by default 1, and the R^2 a regression including the unobservables would reach, by default $\min(1.3R_1, 1)$ with R_1 that of the controlled regression; `rmax()` lies in $(0, 1]$.

`sign(#)` acts on `method(lsz)` and takes 1 or -1 . The moments of LSZ identify γ_1 up to an orientation; the option picks the branch the solver follows, as `trigmm`'s own option does. Keep the branch whose implied parameters are admissible.

`vce(svy)` computes the standard errors by linearization over the design declared by `svyset`; see Section 3.5.

`level(#)` sets the confidence level; `freeiv`, `level(#)` redisplay the last estimates at another level.

`noheader` suppresses the header.

3.3 Which option acts on which route, and what the command refuses

An option that is legal but inert is the same trap as one silently absorbed. `delta()` and `rmax()` act on `oster` alone, `sign()` on `lsz` alone; every other route takes none of them, the default included, and `method(all)` uses all three, each on its own row. With two endogenous regressors, `method()`, `delta()`, `rmax()` and `sign()` are refused rather than ignored.

The command also refuses `sign()` other than ± 1 , `rmax()` outside $(0, 1]$, a missing `delta()`; `vce()` other than `svy`, and `vce(svy)` with weights on the command line, on data not `svyset`, or with a design it does not handle (`singleunit(scaled)` or `centered`, poststratification); a variable in two roles; and `method(lewbel12)`, `method(rank)` or `method(oster)` with no exogenous control. Numerical and statistical failures are reported, not hidden. A route asked for alone that has no estimate on the data ends, after its display and with `e()` posted, with a return code: `r(430)` for an LSZ solver that does not converge, as Stata's own estimators do, and `r(498)` for a route with no value or a guard of model B. `method(all)`, a table of routes, never does.

3.4 How to read the output

The display has four blocks, meant to be read in order.

The coefficient with its standard error and interval when one is valid. When it is not — a point-only comparison route, a minimum on the boundary, a solver that did not converge — the point is printed alone and a note says which of those it is. A suppressed standard error is a statement, not an omission.

Identification the least-squares slope, the interval (5), the skewness of the first-stage residual, the discriminant D with its standard error and a z against zero, the two roots of (6) and how many lie inside the interval.

Nuisances implied by the retained value θ , $\sigma_{V_2}^2$, $\sigma_{V_1}^2$ and c , computed *from the coefficient actually retained*, with A and B . A negative variance here is the loudest signal the command produces: the retained value lies outside the region the model allows, and the numbers are arithmetic rather than estimates.

Over-identification at order four the joint GMM, its J , the region $\mathcal{C}$ and its share of the interval, then the profiled GMM and its J , then the factor that decides whether that J can test anything at all.

3.5 Standard errors, weights, and survey data

The analytic standard errors. The two ends of the interval, `qme`, `hme`, `gmm`, `pgmm`, `lsz` and the closed form of model B have one: the delta method on the covariance of the contributions of the moments, the estimation of the coefficients on X accounted for. `lewb12`, `copula`, `rank` and `oster` return a point only.

Weights. Every moment is weighted. With `aweights` the covariance of the contributions is their weighted mean square. With `pweights` it takes the sandwich form, every observation its own primary unit; with all weights equal to 1 that is the unweighted standard error times $\sqrt{n/(n-1)}$.

Survey data. Under `vce(svy)` the design declared by `svyset` enters the covariance of the contributions: linearized values summed by primary sampling unit, deviations within strata, the finite-population correction (a sampling rate, or the number of units in the stratum), and t on the design degrees of freedom, the number of units less the number of strata. This is the linearization `svy` uses: on the `ols` route it reproduces `svy: regress`. The first stage of the design is used; a multi-stage design with a finite-population correction at the first stage has that correction dropped, which treats the units as drawn with replacement and errs towards a larger variance, and the output says so. A stratum with a single unit makes the standard errors missing under `singleunit(missing)`, as `svy` does, and contributes nothing under `singleunit(certainty)`. The two GMM build their weight matrix from the design's covariance, so where it is missing they return no estimate. `if` and `in` restrict the design to the observations selected; there is no `subpop()`.

3.6 The bootstrap, and what it can and cannot do

Under the `bootstrap` prefix each replication runs the whole route again. The prefix is the inference for the routes with no analytic standard error, and it repairs the delta method where the estimator is a smooth function of the data that the linearization describes poorly. The qme near the edge is the case in point. Its standard error is proportional to $1/\sqrt{D}$, and when the z of D is below 2 the delta method understates its sampling error: on `freeiv_sim1` (Section 4.1, z of D 1.49) it gives 0.044, against 0.053 for 500 replications.

```
. use freeiv_sim1, clear
(freeiv sim1: U chi2(3), V normal, gamma=0.4, alpha2=0.8)

. set seed 20261006

. bootstrap, reps(500) nodots: freeiv y1 x (y2)
```

	Observed	Bootstrap			Normal-based
y1 coefficient	std. err.	z	P> z	[95% conf. interval]	
y2 .4355823	.0527482	8.26	0.000	.3321978	.5389668

```
bootstrap: 500 of 500 replications have an estimate (100%)
(output omitted)
```

The bootstrap is no remedy where the estimator is not smooth: where the two roots of (6) merge, where the third-order signal is weak, where a minimum lies on the boundary of the interval. There the limiting law depends on a local parameter — the distance to the singularity in units of its standard error — which no resampling estimates (Andrews 2000; Andrews and Cheng 2012); and where the parameter may be unidentified, no interval of the form estimate $\pm c \times$ standard error keeps its coverage (Dufour 1997). The interval (5), which rests on the second order alone, is the inference that survives there, with the region $\mathcal{C}$ after it.

A second limit concerns the replications in which a route has no estimate: an LSZ system with no root, an hme whose ratio has the wrong sign, a guard of the two-indicator model. Such a replication ends with a return code (`r(430)` or `r(498)`); the prefix drops it and counts it, and the command prints the share of the replications that have an estimate. The statistics are then those of the draws in which the data meet the conditions of the route's model — a selected subset, whose spread describes the estimator where the model fits, not the uncertainty about γ_1 . When the share is close to one the selection is immaterial; when it is not, the share is the result, and the conditional law cannot be estimated uniformly either (Leeb and Pötscher 2006). A minimum of `gmm` on the boundary is an estimate, and its draw is kept.

On Wooldridge's `wage1` data, the official example of `trigmm`, LSZ has a root on the full sample (0.024, standard error 0.019) but not in most bootstrap draws:

```
. set seed 20260913

. bootstrap, reps(50) nodots: freeiv lwage exper (educ), method(lsz)
-----+-----
          |      Observed      Bootstrap
          | lwage | coefficient  std. err.      z      P>|z|      Normal-based
          |-----+-----|-----+-----|-----+-----|
          |          |          |          |          |          |          |
educ |          | .0244438    .0196602    1.24    0.214    -.0140895    .0629771
          |-----+-----|-----+-----|-----+-----|

bootstrap: 20 of 50 replications have an estimate (40%)
the statistics above are computed on these 20 only (help freeiv, bootstrap)
(output omitted)
```

The 20 draws kept have a spread of 0.020, close to the analytic standard error, yet they centre on 0.041, almost one standard error above the full-sample estimate. Given the same draws and 1,000 iterations, `trigmm` finds no root either in 21 of the 30 others and stops near one in three; in six it finds one that the solver of `freeiv` misses, which would not change the verdict. The route is fragile on these data, and its count, not its standard error, is what to report.

3.7 Stored results

`freeiv` is an `e()`-class command; `ereturn list` shows the full set. In model A the ones most often wanted are `e(gamma)` and `e(se)` for the retained estimate, `e(b)` and `e(V)` for the prefixes and the postestimation commands; `e(lo)`, `e(hi)`, `e(se_lo)` and `e(se_gt)` for the interval; `e(qme)`, `e(vertex)`, `e(at_vertex)` and `e(hme)` with their standard errors; `e(g_gmm)`, `e(J_gmm)`, `e(p_gmm)` for the joint GMM, with `e(ar_lo)`, `e(ar_hi)` and `e(ar_frac)` for the region (7) and `e(gmm_bound)`, `e(gmm_weak)` for its flags; `e(g_pgmm)`, `e(J_pgmm)`, `e(p_pgmm)` for the profiled GMM; `e(idfac)` and `e(z_idfac)` for the factor that decides whether the J can test; `e(disc)` and `e(disc_z)` for the discriminant; `e(theta)`, `e(sV1)`, `e(sV2)`, `e(cshare)`, `e(A)`, `e(B)` for the implied nuisances; `e(lsz)`, `e(lsz_conv)`, `e(lewbel12)`, `e(copula)`, `e(g_rank)`, `e(oster)` for the comparison routes. At a boundary minimum of `method(gmm)`, `e(V)` still carries the delta-method variance that the display suppresses (`e(se)` is missing), so that a bootstrap draw on the boundary is kept. Under `vce(svy)`, `e(df_r)`, `e(N_strata)` and `e(N_psu)` describe the design. In model B, `e(g2)`, `e(g3)` and `e(a1)` are the structural coefficients and the free loading, `e(sc_d)` and `e(z_sc)` the scale-consistency test, `e(q_J)` the J of the GMM on sixteen moments, `e(ivgap)` what the second indicator would return as an instrument, and `e(guard)` which guard fired, or zero.

3.8 Companion commands

`freeivmenu` reports, *before* any estimation, which assumptions the data can carry: the width of the interval, the skewness of the first-stage residual, the z of m_{03} and of the discriminant with the standard error it implies for the quadratic route, how many of its roots lie inside the interval, B for the symmetry route, and a heteroskedasticity test for `lewbel12`. Each line carries a verdict in words. It also prints the *local slope profile*: the kernel-weighted slope of $\hat{\xi}$ on $\hat{\varepsilon}_2$ at seven per-

centiles of $\hat{\varepsilon}_2$, which in the model equals $\gamma_1 + \alpha_1 \partial E[U \mid \hat{\varepsilon}_2] / \partial \hat{\varepsilon}_2$. A flat profile means a Gaussian-like confounder, on which the third order will find little; when $E[U \mid \varepsilon_2]$ is increasing, the minimum of the profile bounds γ_1 from above more tightly than $\tilde{\gamma}$; and the ratio of the two tail slopes tends to 2 under scale consistency. It is a check on the routes, not a route: a curved profile is what a skewed confounder and a nonlinear outcome equation both produce.

With two variables in the parentheses the menu switches to the two-indicator model and reports what its closed form needs: the t of the residual correlation of the two indicators, against the rule $t \geq 10$; the two third-order cross-moments with their z and whether they share a sign, the first guard; the implied loadings and idiosyncratic variances, the two other guards; the implied skewness of U , which sets the precision regime — near 0.5 the causal coefficient is recovered but the free loading a_1 is not, above 1.5 every parameter is precise and scale consistency becomes testable; and the one-factor check. A guard that fires in the menu fires in the command.

freeivdiag takes any value of γ_1 — from **freeiv**, from two-stage least squares, from another paper — and reports what it implies about the unobservables under the maintained model: its position in the interval, the implied variances, the confounder's share, A and B , and whether the value is feasible at all.

freeivtest tests endogeneity itself, with no estimate of γ_1 : under $\theta = 0$ the least-squares residual of $\hat{\xi}$ on $\hat{\varepsilon}_2$ is independent of $\hat{\varepsilon}_2$, so two cross-moments vanish, and their Wald test keeps its power where the qme loses its own. It also tests the agreement between the closed forms and confronts an outside estimate, with its own standard error, with the interval. After model B it reports the scale-consistency test, the one-factor check and the J .

freeivreport assembles all of it in one table, optionally written to a file.

4 Examples

Four datasets travel with the package (**net get freeiv**, or **ssc install freeiv, all**): **freeiv_sim1**, **freeiv_sim2** and **freeiv_proxy** are simulated, with their generating equations printed in the help file, and **freeiv_card** is the Card (1995) extract distributed with Wooldridge's teaching datasets. Every example below is also an example of **help freeiv**, where it runs from its links in three forms that give the same commands: in the command window, in the dialog box filled in, or as a do-file. The outputs are abridged where marked.

4.1 Where the model holds

freeiv_sim1.dta is generated from

$$\begin{aligned} u &= (\chi_3^2 - 3)/\sqrt{6}, & v_1, v_2, x &\sim N(0, 1), \\ y_2 &= 0.30x + 0.80u + v_2, \\ y_1 &= 0.60x + 0.40y_2 + 0.32u + v_1, \end{aligned}$$

so the truth is $\gamma_1 = 0.40$, and scale consistency holds by construction, $0.40 \times 0.80 = 0.32$. The menu comes first.

```
. use freeiv_sim1, clear
(freeiv sim1: U chi2(3), V normal, gamma=0.4, alpha2=0.8)

. freeivmenu y1 x (y2)
```

Identification menu: what these data can carry
y1 on y2, controls x

route	signal	value	verdict
bounds	width of the interval	0.2823	always
	[gamma-tilde/2, gamma-tilde]	[0.2823, 0.5646]	
third order: qme,	skewness of eps2	0.3879	
hme, lsz	z of m03	5.68	strong
	z of the discriminant D	1.49	weak
	implied s.e. of the QME	0.0436	usable
	A = alpha2^3 E[U^3]	0.8195	
	mu = A/(A+B)	0.9742	
symmetry of V2: hme	B = E[V2^3], hme sets 0	0.0217	plausible
heteroskedasticity:	F of eps2^2 on X	0.78	absent
lewbel12	p	0.3773	
local slope profile	slope of xi on eps2 at p5	0.4703	h = 0.500
of E[xi eps2]	p10 / p25 / p50	0.4694 0.4882 0.5481	
	p75 / p90 / p95	0.5658 0.6177 0.6926	
	curvature z (max-min)	2.80	curved
	min slope, at p10	0.4694	tightens
	tail ratio p95/p5	1.47	not 2
two indicators	freeivmenu y1 x (y2 y3)	.	by syntax

(output omitted)

The third-order signal is strong, the discriminant weakly positive, and the symmetry of V_2 plausible; heteroskedasticity is absent, so `lewbel12` has nothing to work with. Every route, against the interval:

```
. freeiv y1 x (y2), method(all)
(output omitted)
Estimates of gamma      estimate   std. err.
the interval
  bounds                [0.282288, 0.564577]
  ols = gamma-tilde     0.564577    0.011820
order 3
  qme                   0.435582    0.043598
  hme                   0.448740    0.021985
order 4
```

```

      gmm                0.445484      0.038063  J 0.143  p 0.705
      pgmm               0.445316      0.037618  J 0.143  p 0.706
other maintained models
      lsz                0.563742      .         did not converge
      lewbel12           1.384843      .         outside the bounds
      copula             1.353432      .         outside the bounds
      rank               1.350008      .         outside the bounds
      oster              0.451458      .
(output omitted)
Over-identification at order four: nine moments, eight parameters
joint GMM, method(gmm)
      gamma              0.445484
      J, 1 df            0.1432      P>chi2 0.7051
      J within 3.84 of its minimum on [0.372856, 0.555166]
      that is 64.6% of [gamma-tilde/2, gamma-tilde]
(output omitted)

```

The routes of the maintained model agree with one another and with the truth: 0.436 for the qme, 0.449 for the hme, 0.445 for both GMM, each inside the interval $[0.282, 0.565]$. The region covers 65% of the interval: the moments of orders three and four narrow what the second order gives. The two GMM agree to the third decimal, so the criterion is not flat. `lewbel12`, `copula` and `rank` land far outside the interval: their own assumptions do not hold in this design, and the interval shows it, before any knowledge of the truth.

4.2 The edge of identification

`freeiv_sim2.dta` differs in one respect: v_2 is drawn from the same skewed law as u . Scale consistency still holds and the truth is still 0.40, but now $B = E[V_2^3] = 1.633$ nearly equals $2A = 2 \times 0.8^3 \times 1.633 = 1.672$, so the population discriminant is $\gamma_1^2(2A - B)^2 = 0.0002$, against 0.45 in `freeiv_sim1`. The design sits almost exactly on the surface where the two roots merge.

```

. use freeiv_sim2, clear
(freeiv sim2: U and V2 chi2(3), gamma=0.4, alpha2=0.8)

. freeiv y1 x (y2), method(all)
(output omitted)
order 3
      qme                0.419948      0.017144  vertex
      hme                0.335688      0.015063
order 4
      gmm                0.409801      0.218756  J 2.016  p 0.156
      pgmm               0.402335      0.133679  J 1.987  p 0.159
(output omitted)
      qme: negative discriminant, no real root -- the vertex
      3 m12/(4 m03), where the two roots merge
(output omitted)
      discriminant D = 9m12^2 - 8m03m21      -0.538322 (s.e. 0.631877)
      D = gamma^2 (2A - B)^2 : z against 0 -0.851940
(output omitted)

```

In this sample D is slightly negative, within its noise ($z = -0.85$): the two roots merge, and the qme returns the vertex, 0.420, and says so. The hme returns 0.336 with

a standard error of 0.015 — and is wrong, because the symmetry it assumes is violated by construction; the output reads B at the vertex, 1.49, far from zero. A route that fails loudly is safer than one that fails quietly. The GMM routes still answer, at the price of their precision: 0.410 and 0.402, with standard errors of 0.22 and 0.13.

4.3 Real data with one regressor: the interval is the result

`freeiv_card.dta` is the Card (1995) extract used in Araar (2026d): log wage on schooling, with experience, its square, and indicators for race, region and metropolitan status. The menu comes first.

```
. use freeiv_card, clear

. freeivmenu lwage exper expersq black south smsa (educ)

Identification menu: what these data can carry
lwage on educ, controls exper expersq black south smsa
n = 3010
```

route	signal	value	verdict
bounds	width of the interval	0.0370	always
	$[\gamma\text{-tilde}/2, \gamma\text{-tilde}]$	[0.0370, 0.0740]	
third order: qme,	skewness of eps2	0.1715	
hme, lsz	z of m03	2.43	weak
	z of the discriminant D	0.01	absent
	implied s.e. of the QME	0.2537	wide
	$A = \alpha^2 E[U^3]$	0.5850	
	$\mu = A/(A+B)$	0.4629	
symmetry of V2: hme	$B = E[V^2^3]$, hme sets 0	0.6788	doubtful
heteroskedasticity:	F of eps2^2 on X	19.52	present
lewbel12	p	0.0000	

(output omitted)

The verdicts are discouraging for a point before any estimate is formed: the skewness of the first-stage residual is 0.17, the z of m_{03} weak, the z of D absent, and the qme's implied standard error is several times the width of the interval.

```
. freeiv lwage exper expersq black south smsa (educ), method(all)
(output omitted)
the interval
  bounds      [0.037004, 0.074009]
  ols = gamma-tilde 0.074009 0.003638
order 3
  qme      0.072146 0.253687
  hme      .
order 4
  gmm      0.074008 . J 2.675 p 0.102
  pgmm     0.073909 0.060269 J 2.681 p 0.102
other maintained models
  lsz      0.074091 . did not converge
  lewbel12 0.075721 . outside the bounds
  copula   0.008638 . outside the bounds
  rank     0.057380 .
  oster    0.083968 . outside the bounds
```

```
(output omitted)
J within 3.84 of its minimum on [0.037005, 0.074008]
that is 100.0% of [gamma-tilde/2, gamma-tilde]
note: that region is the WHOLE identified interval, so the
moments of orders 3 and 4 add nothing here to the
assumption-free bounds. The point estimate is the
argmin of a flat criterion; report the interval.
(output omitted)
```

On Card the data support the interval, 0.037 to 0.074. The qme, 0.072, is reported with its precision, a standard error of 0.25, and the package's diagnostics say themselves that the third-order signal is weak: the menu before estimation, and the region of the joint GMM, which covers the whole interval, with its minimum on the boundary. Card is the example where the qme itself signals that its point should not be read — a strength of the method, not a weakness; `freeiv.sim1` remains the example where it works. Three of the comparison routes fall outside the interval. That is not a verdict on those estimators: their maintained models and this one disagree about these data, and under this one those values imply a negative variance.

With fewer controls the qme moves, within its noise, outside the interval:

```
. freeiv lwage exper expersq (educ)
(output omitted)
gamma (educ)      0.119280  s.e. 0.039585  [ 0.041695,  0.196865]
z =      3.013      P>|z| = 0.0026
note: the estimate lies OUTSIDE the identification bounds:
      under scale consistency it implies a negative variance
      (see the nuisances below, or -freeivdiag-)
(output omitted)
bounds, c in [0, 1]                                [0.046585, 0.093171]
(output omitted)

. freeivdiag
(output omitted)
Implied nuisances (second-order system, functions of gamma alone)
theta = alpha2^2 Var(U)                             -0.899798  NEGATIVE
sigma2_V2                                           5.010482
sigma2_V1                                           0.173993
c = theta/(theta + sigma2_V2)                       -0.2189   outside [0, 1]
(output omitted)
```

The root, 0.119, lies above the interval [0.047, 0.093], though not significantly given its standard error of 0.040, and `freeivdiag` shows the negative variance such a value implies. Removing controls also changes the model itself — the least-squares slope goes from 0.074 to 0.093 — so the move is not noise alone.

4.4 An estimate from elsewhere, against the interval

Card's instrumental-variable estimate with college proximity is 0.132 (standard error 0.049), above the least-squares slope. Under scale consistency least squares can only overstate γ_1 , so the two are incompatible — but at what precision?

```
. quietly freeiv lwage exper expersq black south smsa (educ)

. freeivtest, gamma(0.132) segamma(0.049)
(output omitted)
Endogeneity itself: theta = 0, with no estimate of gamma
E[r eps2^2] = m12 - gt m03 0.039853
E[r^2 eps2] = m21 - 2gt m12 + gt^2 m03 0.002892
Wald, chi2(2) 1.4271 P>chi2 0.4899
(output omitted)
Outside estimate 0.132000 against the identified set
bounds [0.037004, 0.074009] s.e. 0.001819 and 0.003638
distance to the upper bound 0.057991
z 1.1802 P>|z| 0.2379
(output omitted)
```

The instrument lies 0.058 above the upper end, with $z = 1.18$: outside the set, not significantly. Choosing between the instrument and scale consistency is the researcher's call; these data do not make it. The test of endogeneity itself does not reject either ($p = 0.49$): detecting $\theta > 0$ needs a third-order signal in the first-stage residual, and on Card that signal is weak.

4.5 A second indicator of the confounder

Mother's education loads on the family background that confounds schooling, and may affect wages itself: it is an indicator, not an instrument. The menu says, before anything is estimated, what the pair can carry.

```
. freeivmenu lwage exper expersq black south smsa (educ motheduc)
(output omitted)
relevance of the pair corr(eps2, eps3), t 14.61 strong
alpha2 alpha3 = E[eps2 eps3] 1.5245 z 12.59
rule of Araar (2026d): t >= 10
third order E[eps2^2 eps3], z 1.15 absent
E[eps2 eps3^2], z 1.71 absent
same sign: guard (i) yes passes
(output omitted)
confounder implied skewness E[U^3] 0.4666 low
g2 recovered, a1 imprecise
(output omitted)
```

The pair is relevant ($t = 14.6$) and the first guard passes, but the two third-order cross-moments are weak and the implied skewness of the confounder is 0.47: the regime in which γ_2 is recovered but a_1 is imprecise. That is the precision the estimates have.

```
. freeiv lwage exper expersq black south smsa (educ motheduc)
(output omitted)
Structural coefficients
g2 (educ) 0.065666 s.e. 0.016321 [ 0.03368, 0.09765]
g3 (motheduc) -0.009110 s.e. 0.031368 [-0.07059, 0.05237]
a1 (U in the outcome) 0.066575 s.e. 0.137635 [-0.20318, 0.33633]
(output omitted)
Scale consistency, which model A cannot test
a1 (free, identified here) 0.066575
g2 a2 + g3 a3 0.034221
difference 0.032354 s.e. 0.214934 z 0.151
```

```
(output omitted)
What motheduc used as an instrument for educ would return
iv estimate                                0.101543
(output omitted)
```

Now the coefficient on schooling is a point, 0.066 with a standard error of 0.016, and the confounder's direct effect $a_1 = 0.067$ is estimated rather than assumed. Scale consistency is tested: the departure 0.032 has a standard error of 0.21 — not rejected, and, at this skewness of the confounder, not informative either. Used as an instrument, mother's education would return 0.102, 0.036 above γ_2 . These numbers reproduce Table 6b of Araar (2026d); with father's education the coefficient is 0.035, with a departure from scale consistency of 0.563.

Two other indicators of the file are refused by the guards, which is the intended behaviour. The command then ends with `r(498)`, so a do-file runs it under `capture`:

```
. capture noisily freeiv lwage exper expersq black south smsa (educ IQ)
(output omitted)
no estimate: guard 3 of Proposition 1 fired
    an implied idiosyncratic variance is negative: the common
    factor would have to explain more than the whole variance of
    an indicator
-----

. display "return code " _rc ", guard " e(guard)
return code 498, guard 3
```

IQ implies a negative idiosyncratic variance (guard 3), KWW third-order cross-moments of opposite signs (guard 1): reason by reason, Table 6a of Araar (2026d). The menu announces both refusals before the command is run.

4.6 A model the data refute

The third-order signal can be strong and the model still wrong. On the `nls88` data shipped with Stata, with tenure as the endogenous regressor:

```
. sysuse nls88, clear
(NLSW, 1988 extract)

. generate lwage = ln(wage)

. freeiv lwage grade age i.race (tenure)
(output omitted)
gamma (tenure)    0.045892   s.e. 0.003080   [ 0.039856,  0.051928]
z = 14.902      P>|z| = 0.0000
note: the estimate lies OUTSIDE the identification bounds:
      under scale consistency it implies a negative variance
      (see the nuisances below, or -freeivdiag-)
(output omitted)
bounds, c in [0, 1]                                [0.013677, 0.027353]
skewness of the first-stage residual                1.001465
(output omitted)
D = gamma^2 (2A - B)^2 : z against 0 5.655590
roots of the quadratic                            -0.015164 and 0.045892
```

```

of which inside the bounds          0
(output omitted)
profiled GMM, method(pgmm): orders 2 and 3 fitted exactly
gamma                                0.027253
J, 1 df                             26.6722   P>chi2 0.0000
(output omitted)

```

The signal is strong — a skewness of 1.00, a z of D of 5.66 — and the qme is precise, 0.046 with a standard error of 0.003. But neither root lies inside the interval $[0.014, 0.027]$, the implied confounder variance is negative, and the J at order four rejects (26.7 for the profiled GMM, 34.4 for the joint one). A precise estimate is not a valid one: the one-factor model with scale consistency does not describe tenure and wages in these data, and the output says so.

4.7 Survey data

The NHANES II data of the Stata manuals carry a design — strata, primary sampling units, sampling weights — and `vce(svy)` uses it:

```

. webuse nhanes2, clear

. freeiv bpsystol age female black (bmi), vce(svy)

Instrument-free estimation      Number of obs   =   10,351
                               Number of strata =     31
                               Number of PSUs   =     62
                               Design df       =     31

model: bpsystol on bmi, controls age female black
method: qme
(output omitted)
discriminant D = 9m12^2 - 8m03m21      -1.576e+05 (s.e. 6.428e+04)
D = gamma^2 (2A - B)^2 : z against 0 -2.452221
note: D is NEGATIVE beyond its sampling noise. Under the model
      D = gamma^2 (2A - B)^2 cannot be below zero, so these data
      reject the linear one-factor model with scale consistency,
      and the vertex below is not an estimate.
(output omitted)

```

The design has 31 strata and 62 primary units, hence 31 degrees of freedom and intervals on Student's t . The standard error of the least-squares slope is 0.050 under the design, against 0.047 with the sampling weights alone, which ignore the clustering. And the diagnostics speak: the discriminant is negative beyond its noise ($z = -2.45$), which no one-factor model can produce, and the J rejects (9.5 for the profiled GMM, 8.5 for the joint one). Body-mass index and blood pressure in NHANES II do not fit this model.

4.8 Reporting a range, and factor variables

Oster's adjustment depends on what is assumed. On Card, with the controls of Section 4.3, $R_{\max} = 0.5$ gives 0.098, and $\delta = 2$ with $R_{\max} = 0.8$ gives 0.190. Both lie above the least-squares slope, 0.074, and so outside the interval: on these data, proportional

selection of that size and scale consistency cannot both hold. Factor variables among the controls give what hand-made terms give: with `c.exper##c.exper i.black i.south i.msma` the qme differs from the one of Section 4.3 by less than 10^{-12} .

5 Discussion

Three habits are worth stating, because they are what the command is designed to support.

Read the interval before the point. It costs no distributional assumption and cannot fail unless the model itself is wrong. Everything else in the output is an attempt to narrow it, and the attempt can fail.

Read the diagnostic before the estimate. `freeivmenu` says which assumptions the data carry; the discriminant says whether the quadratic route has anything to work with; the share of the interval covered by (7) says whether the fourth-order moments added anything. When those say no, the point estimate is an artefact of a flat criterion, and the interval is what to report.

Read the routes together. Agreement between estimators with different maintained models is evidence; disagreement locates the assumption that is doing the work. This is why `method(all)` exists and why the comparison routes are computed on the same data rather than cited.

The limitations are equally plain. Model A has one latent confounder and is linear in it, and it maintains scale consistency; the two-indicator model frees the direct loading and tests that restriction, but it keeps the single factor, and a second factor that is Gaussian is undetectable at any order of moments. The J tests scale consistency and linearity jointly and cannot separate them, and an interaction between the regressor and the confounder, which the model leaves out, is not among the departures it is built to detect.

What the third order cannot tell apart. One caveat deserves its own paragraph, because it concerns every instrument-free route and not only these. In the single-indicator model the three third-order moments serve three unknowns, (γ_1, A, B) : the system is just-identified, so any triple (m_{03}, m_{12}, m_{21}) can be rationalized by a confounder, and a mechanism with no confounder at all can produce the same triple. Take $Y_1 = aY_2 + bY_2^2 + e$ with Y_2 standard normal, e independent, and no latent factor. Then $m_{03} = 0$, $m_{12} = 2b$, $m_{21} = 4ab$, and (6) reduces to $-6b\gamma_1 + 4ab = 0$: the route returns $\gamma_1 = 2a/3$ for every $b \neq 0$ — a correction of one third of the least-squares slope that does not depend on the size of the quadratic term, lies inside (5), carries a positive discriminant, and implies a confounder share $c = 1/2$ with positive variances. Nothing at order three can object, because nothing at order three is over-identified. The fourth order can: on this design with $a = 1$, $b = 0.1$ and $n = 20,000$, the qme is 0.684 and the J of `method(pgmm)` and `method(gmm)` are 36.1 and 37.4, both rejecting. But the power of that test falls with b and with the sample size, while the bias does not: on the same draws with $b = 0.02$, the two J fall to 4.0, at the edge of rejection ($p = 0.046$ and 0.044),

and the qme is still 0.678. Three things follow. The z of the discriminant in `freeivmenu` measures the strength of a third-order signal, not its source — a curvature in $E[\hat{\xi} \mid \hat{\varepsilon}_2]$ is what a skewed confounder and a nonlinear outcome equation both produce, and a RESET-type check on $\hat{\varepsilon}_2^2$ rejects under either. The two-indicator model is the least exposed route in the family: its loading layer never reads Y_1 , so a nonlinearity of the outcome equation can reach only the causal layer. And a valid external instrument, where one exists, does not rest on the third-order structure at all: an instrument-free route complements one, it does not replace it.

6 Acknowledgments

The reference implementations `ivreg2h`, `trigmm`, and `psacalc` were used to validate the comparison routes. This note documents version 1.0.0 of the package and is archived at <https://doi.org/10.5281/zenodo.22770175>; every output and number in it is reproduced by `replication/freeiv_paper/freeiv_note.do` in the repository of the package. It is written in the style of a Stata Journal article and is not a journal submission. Stata is a registered trademark of StataCorp LLC.

7 References

- Andrews, D. W. K. 2000. Inconsistency of the bootstrap when a parameter is on the boundary of the parameter space. *Econometrica* 68: 399–405.
- Andrews, D. W. K., and X. Cheng. 2012. Estimation and inference with weak, semi-strong, and strong identification. *Econometrica* 80: 2153–2211.
- Araar, A. 2026a. Correcting endogeneity without external instruments: Four closed-form estimators for the linear structural model. Working paper, Zenodo. <https://doi.org/10.5281/zenodo.20312356>.
- . 2026b. The quadratic moment estimator for instrument-free endogeneity correction. Working paper, Zenodo. <https://doi.org/10.5281/zenodo.22068003>.
- . 2026c. Instrument-free estimation under linear scale-consistency: Closed-form identification, partial bounds, and higher-moment GMM. Working paper, Zenodo. <https://doi.org/10.5281/zenodo.22119231>.
- . 2026d. Two indicators of one latent confounder: Closed-form identification of the triangular model with a free proxy effect. Working paper, Zenodo. <https://doi.org/10.5281/zenodo.22207331>.
- Baum, C. F., and M. E. Schaffer. 2012. `ivreg2h`: Stata module to perform instrumental variables estimation using heteroskedasticity-based instruments. Statistical Software Components S457555, Boston College Department of Economics.
- Breitung, J., A. Mayer, and D. Wied. 2024. Asymptotic properties of endogeneity corrections using nonlinear transformations. *Econometrics Journal* 27: 362–383.

- Card, D. 1995. Using geographic variation in college proximity to estimate the return to schooling. In *Aspects of Labour Market Behaviour: Essays in Honour of John Vanderkamp*, ed. L. N. Christofides, E. K. Grant, and R. Swidinsky, 201–222. Toronto: University of Toronto Press.
- Dufour, J.-M. 1997. Some impossibility theorems in econometrics with applications to structural and dynamic models. *Econometrica* 65: 1365–1387.
- Imbens, G. W., and C. F. Manski. 2004. Confidence intervals for partially identified parameters. *Econometrica* 72: 1845–1857.
- Kuroki, M., and J. Pearl. 2014. Measurement bias and effect restoration in causal inference. *Biometrika* 101: 423–437.
- Lee, H., A. Lewbel, S. M. Schennach, and L. Zhang. 2026. Instrument-free estimation of triangular equation systems with the trigmm command. *Stata Journal* 26: 68–89.
- Leeb, H., and B. M. Pötscher. 2006. Can one estimate the conditional distribution of post-model-selection estimators? *Annals of Statistics* 34: 2554–2591.
- Lewbel, A. 2012. Using heteroscedasticity to identify and estimate mismeasured and endogenous regressor models. *Journal of Business and Economic Statistics* 30: 67–80.
- Lewbel, A., S. M. Schennach, and L. Zhang. 2024. Identification of a triangular two equation system without instruments. *Journal of Business and Economic Statistics* 42: 14–25.
- Miao, W., Z. Geng, and E. J. Tchetgen Tchetgen. 2018. Identifying causal effects with proxy variables of an unmeasured confounder. *Biometrika* 105: 987–993.
- Oster, E. 2013. psacalc: Stata module to calculate treatment effects and relative degree of selection under proportional selection of observables and unobservables. Statistical Software Components S457677, Boston College Department of Economics.
- . 2019. Unobservable selection and coefficient stability: Theory and evidence. *Journal of Business and Economic Statistics* 37: 187–204.
- Park, S., and S. Gupta. 2012. Handling endogenous regressors by joint estimation using copulas. *Marketing Science* 31: 567–586.
- Stock, J. H., and J. H. Wright. 2000. GMM with weak identification. *Econometrica* 68: 1055–1096.

About the author

Abdelkrim Araar is a researcher at Université Laval and with the Partnership for Economic Policy. His work is in poverty and inequality measurement, development economics, and econometric methodology, and he is the author of several Stata modules in those areas.